

Sri Lanka Institute of Information Technology



BSc (Hons) in Information Technology

Specializing in Data Science

IT3021 – Data Warehousing and Business Intelligence

Year 3

Assignment 2

Semester 1, 2026

Magedaragama M M – IT23837980

Table of Contents

1. Introduction

2. Data Source for Assignment 02

- 2.1 Overview of the Data Source
- 2.2 Data Warehouse Description
- 2.3 Fact Table – FactVehicleAccident
- 2.4 Dimension Tables
- 2.5 ER Diagram
- 2.6 Design Summary

3. SSAS Cube Implementation

- 3.1 Introduction to OLAP Cube
- 3.2 SSAS Project Creation
- 3.3 Data Source Configuration
- 3.4 Data Source View (DSV)
- 3.5 Cube Design
- 3.6 Measures and Fact Table Integration
- 3.7 Dimension Configuration
- 3.8 Hierarchy Implementation
- 3.9 Cube Deployment and Processing
- 3.10 Cube Browsing and Validation

4. OLAP Operations using Excel

- 4.1 Introduction to Excel OLAP Analysis
- 4.2 Connecting Excel to SSAS Cube
- 4.3 Roll-Up Operation
- 4.4 Drill-Down Operation
- 4.5 Slice Operation
- 4.6 Dice Operation
- 4.7 Pivot Operation
- 4.8 Summary of OLAP Analysis

5. Power BI Reports Development

- 5.1 Introduction to Power BI
- 5.2 Data Connection and Preparation
- 5.3 Data Modeling and Relationships

- 5.4 Report 1 – Matrix Report
- 5.5 Report 2 – Slicer-Based Report (Cascading Filters)
- 5.6 Report 3 – Drill-Down Report
- 5.7 Report 4 – Drill-Through Report
- 5.8 DAX Calculations (if used)
- 5.9 Report Publishing (Power BI Service)
- 5.10 Summary of Reports

6. Conclusion

1. Introduction

In modern data-driven environments, organizations rely heavily on analytical systems to extract meaningful insights from large volumes of data. While operational systems efficiently handle transactional data, they are not optimized for complex analysis and reporting. Data Warehousing and Business Intelligence (DWBI) solutions address this gap by transforming structured data into formats suitable for advanced analytics and decision-making.

This assignment builds upon the data warehouse developed in Assignment 01, which was designed using the Road Safety dataset. The data warehouse integrates data from multiple heterogeneous sources, including SQL Server databases, CSV files, and Excel files, and organizes it into a dimensional model consisting of a central fact table and several dimension tables.

The primary objective of this assignment is to utilize the existing data warehouse as a data source for analytical processing and reporting. This is achieved through the implementation of an **OLAP (Online Analytical Processing) cube using SQL Server Analysis Services (SSAS)**, enabling multidimensional analysis of road accident data. The cube is designed by defining measures, dimensions, and hierarchies to support efficient querying and data exploration.

Furthermore, the assignment demonstrates the application of key OLAP operations such as **roll-up, drill-down, slice, dice, and pivot** using Microsoft Excel. These operations allow users to analyze data from multiple perspectives and uncover hidden patterns and trends.

In addition to OLAP analysis, the assignment includes the development of interactive and visually rich dashboards using **Microsoft Power BI**. These reports provide insights into accident trends, environmental conditions, and vehicle-related factors through advanced visualizations, filters, and drill-through capabilities.

Overall, this assignment highlights the transition from data warehousing to business intelligence, demonstrating how structured data can be transformed into actionable insights through analytical tools and visualization techniques. It showcases the practical implementation of end-to-end BI solutions, supporting informed decision-making in the context of road safety and transportation analysis.

2. Data Source for Assignment 02

2.1 Overview of the Data Source

The data source used for Assignment 02 is the **Road Safety Data Warehouse**, which was designed and implemented in Assignment 01. This data warehouse integrates accident and vehicle-related data from multiple heterogeneous sources and transforms them into a structured format suitable for analytical processing.

Unlike traditional OLTP systems, which are optimized for transactional operations, the data warehouse is designed to support **analytical queries and business intelligence applications**. The data is organized using a dimensional modeling approach, enabling efficient aggregation, filtering, and multidimensional analysis.

The data warehouse is implemented in Microsoft SQL Server under the database:

RoadSafety_DW_ETL

This database contains cleaned, integrated, and transformed data, making it an ideal source for SSAS cube creation, Excel-based OLAP analysis, and Power BI reporting required in Assignment 02.

2.2 Data Warehouse Description

The Road Safety Data Warehouse is designed using a **star schema**, which is a widely used dimensional modeling technique in data warehousing. In this design, a central fact table is surrounded by multiple dimension tables.

The purpose of this structure is to:

- Simplify complex queries
- Improve performance for analytical workloads
- Support OLAP operations such as drill-down, roll-up, slice, and dice

The warehouse consists of:

Fact Table

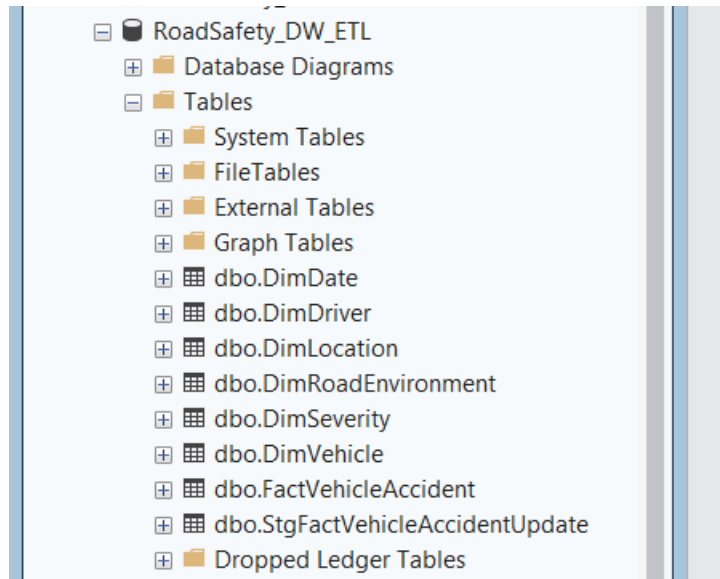
- FactVehicleAccident

Dimension Tables

- DimDate
- DimLocation
- DimRoadEnvironment

- DimVehicle
- DimDriver
- DimSeverity

Each dimension provides a specific analytical perspective, allowing users to analyze accident data based on time, location, environment, vehicle characteristics, driver details, and severity.



Tables in RoadSafety Data Warehouse

This figure shows the fact and dimension tables in the data warehouse.

2.3 Fact Table – FactVehicleAccident

The **FactVehicleAccident** table is the central table in the data warehouse and stores measurable data related to road accidents. Each record represents a vehicle involved in a specific accident, providing a detailed level of analysis.

Key Structure

Primary Key

- FactKey

Foreign Keys

- DateKey
- LocationKey
- RoadEnvKey
- VehicleKey

- DriverKey
- SeverityKey

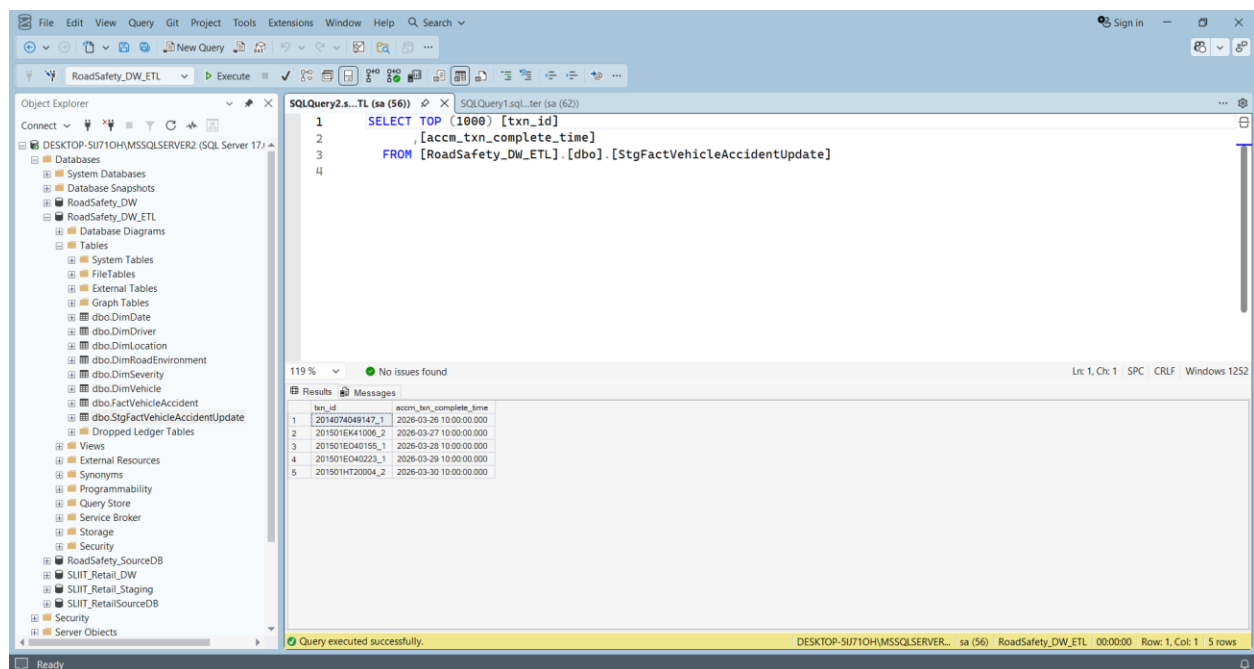
Measures

- Number_of_Casualties
- Number_of_Vehicles
- Age_of_Vehicle
- Engine_Capacity_CC

Transaction Attributes

- txn_id
- accm_txn_create_time
- accm_txn_complete_time
- txn_process_time_hours

These attributes allow users to perform analysis on accident trends, vehicle involvement, and processing times across multiple dimensions.



FactVehicleAccident Table Data

Description: This figure shows sample records from the fact table.

2.4 Dimension Tables

The dimension tables provide descriptive attributes that give context to the measures stored in the fact table. These tables support filtering, grouping, and hierarchical analysis.

DimDate

The DimDate table stores time-related attributes such as year, month, and day, enabling time-based analysis of accidents.

DimLocation

The DimLocation table stores geographical information such as district, region, and country, allowing spatial analysis of accidents.

DimRoadEnvironment

The DimRoadEnvironment table contains environmental conditions such as weather, lighting, and road surface, which influence accident occurrences.

DimVehicle

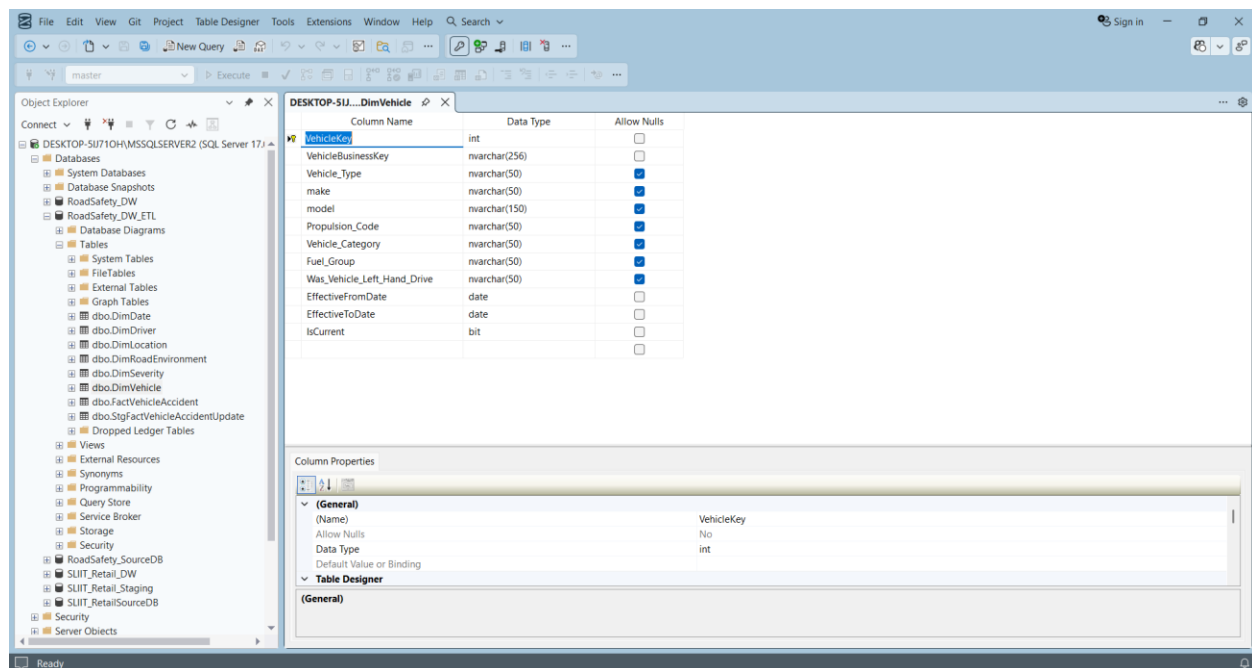
The DimVehicle table stores vehicle-related attributes such as type, model, and fuel type. It is implemented as a **Slowly Changing Dimension (Type 2)** to track historical changes in vehicle data.

DimDriver

The DimDriver table stores driver-related attributes such as age group, gender, and driving characteristics.

DimSeverity

The DimSeverity table stores accident severity levels such as slight, serious, and fatal.



Example Dimension Table (DimVehicle)

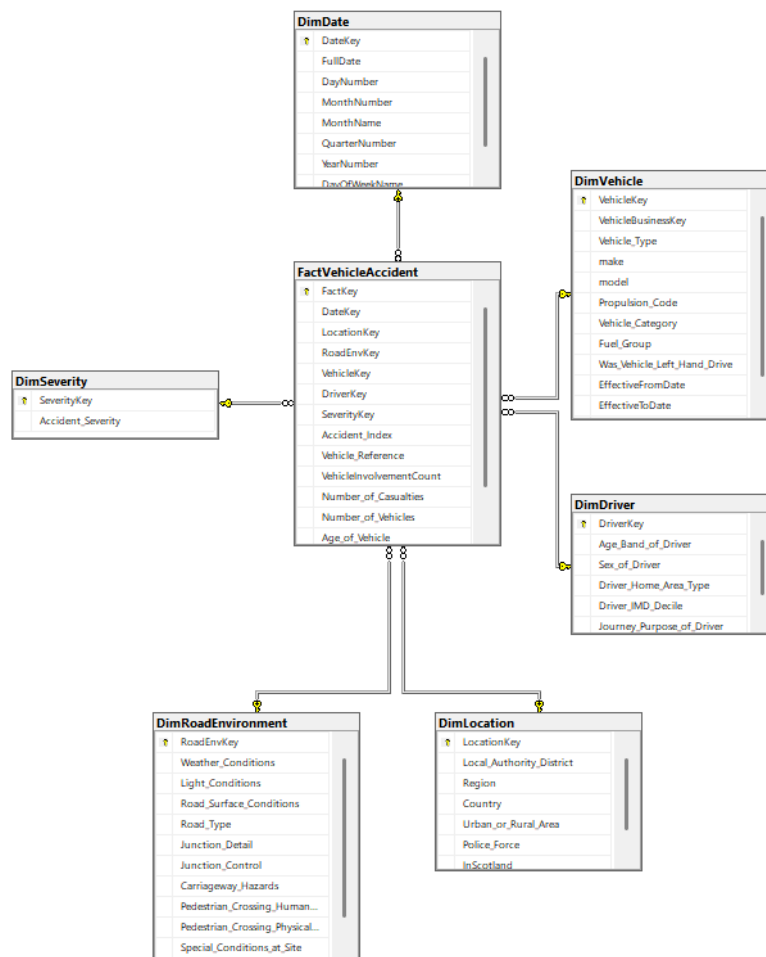
Description: This figure shows the structure and attributes of a dimension table.

2.5 ER Diagram

The following diagram represents the **Entity–Relationship (ER) model of the RoadSafety_DW_ETL data warehouse**, which follows a star schema design. In this model, the central fact table is connected to multiple dimension tables through foreign key relationships.

Each dimension table has a **one-to-many (1:M)** relationship with the fact table, meaning that a single dimension record can be associated with multiple fact records, while each fact record references exactly one record from each dimension.

This structure enables efficient analytical processing and supports multidimensional queries required for business intelligence applications.



ER Diagram of the RoadSafety Data Warehouse (Star Schema)

Description: This diagram illustrates the relationships between the fact table and dimension tables.

2.6 Design Summary

The Road Safety Data Warehouse provides a well-structured and integrated data source for Assignment 02. The star schema design simplifies data access and improves query performance, making it suitable for analytical processing.

The fact table captures measurable accident data, while the dimension tables provide descriptive context for analysis. The one-to-many relationships between dimensions and the fact table enable efficient slicing, dicing, and drill-down operations.

Overall, the data warehouse serves as a strong foundation for implementing SSAS cubes, performing OLAP analysis in Excel, and developing interactive dashboards in Power BI, thereby supporting data-driven decision-making in the context of road safety.

3. SSAS Cube Implementation

3.1 Introduction to OLAP Cube

An OLAP (Online Analytical Processing) cube is a multidimensional data structure designed to enable fast and efficient analytical analysis of large datasets. It allows users to analyze data across multiple dimensions such as time, location, vehicle, and accident severity while supporting operations such as roll-up, drill-down, slice, dice, and pivot.

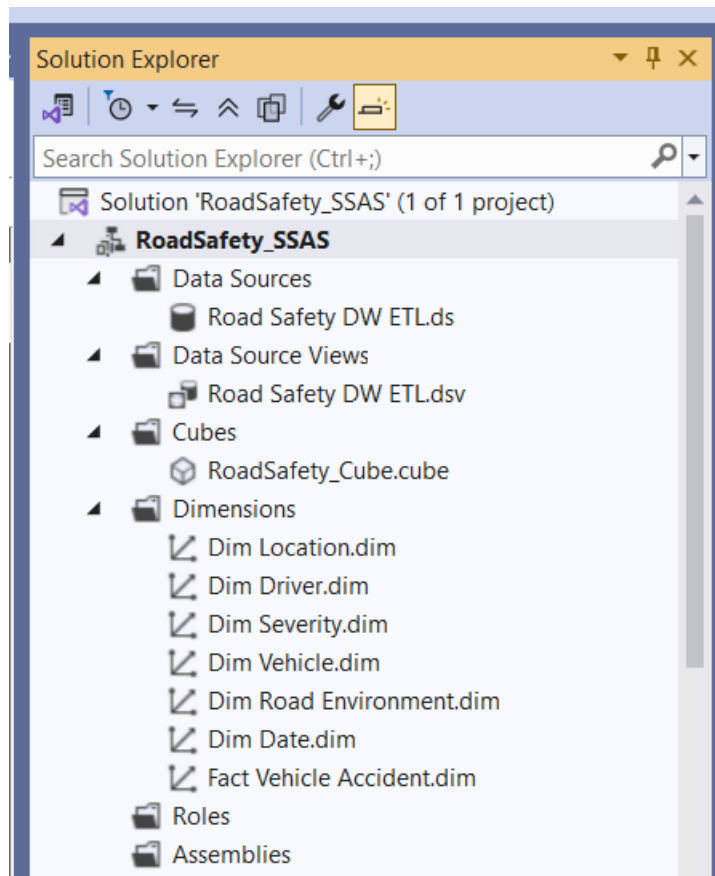
In this project, an OLAP cube was developed using SQL Server Analysis Services (SSAS) based on the RoadSafety_DW data warehouse to support comprehensive analysis of road accident data.

3.2 SSAS Project Creation

An SSAS Multidimensional project named **RoadSafety_SSAS** was created using SQL Server Data Tools (SSDT). This project includes:

- Data Sources
- Data Source Views (DSV)
- Dimensions
- Cube

This structure provides the foundation for building and managing the OLAP cube.



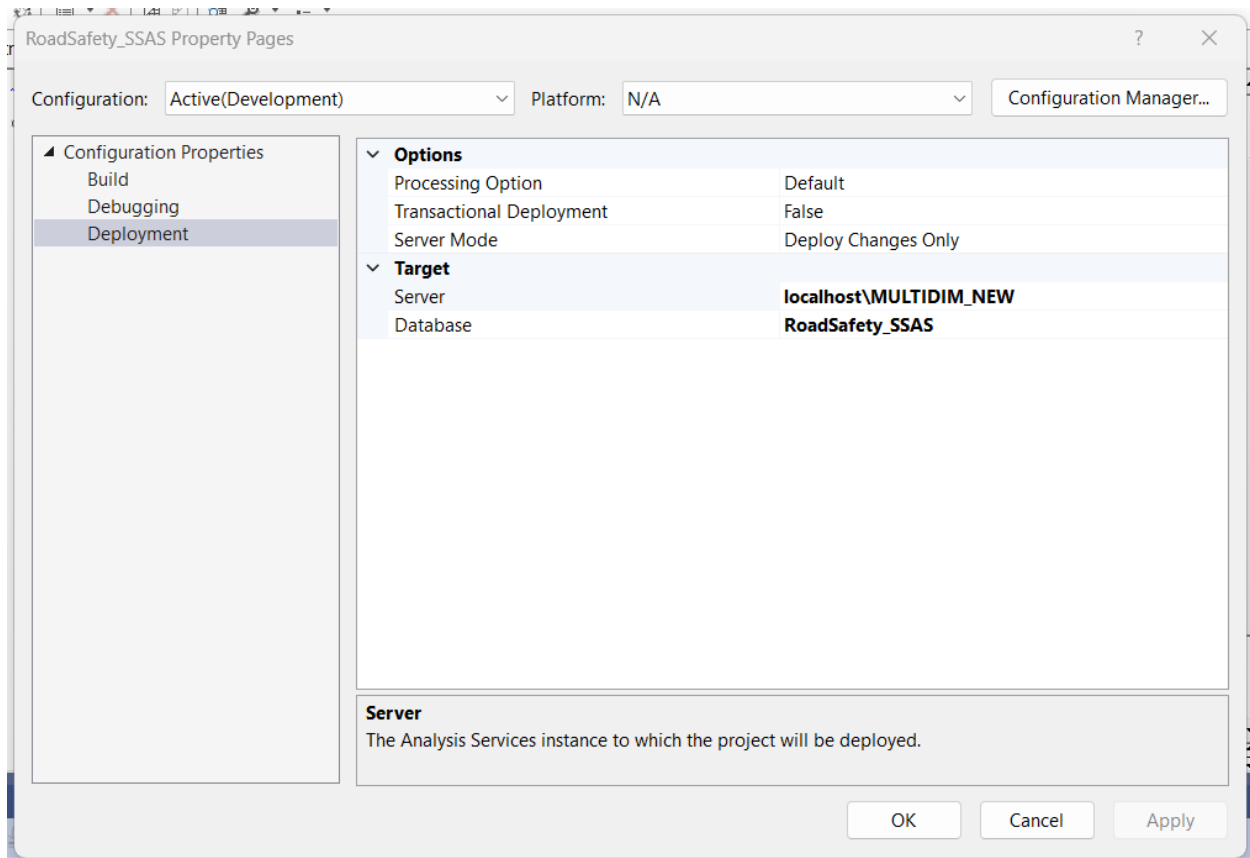
Solution Explorer

3.3 Data Source Configuration

A data source was configured to connect the SSAS project to the **RoadSafety_DW** database. Windows Authentication was used to ensure secure access.

The following configurations were applied:

- Server: localhost\MULTIDIM_NEW
- Database: RoadSafety_DW
- Authentication: Windows Authentication



Data Source Configuration

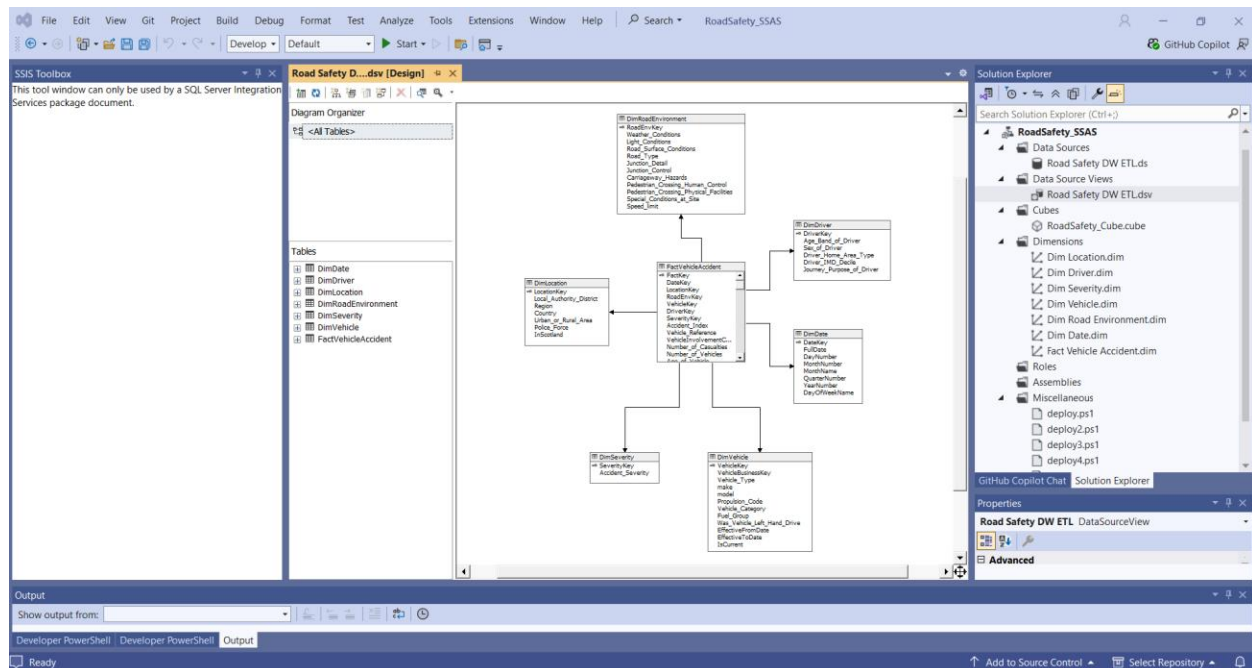
3.4 Data Source View (DSV)

The Data Source View (DSV) was created by selecting the fact table and all related dimension tables. The system automatically detected relationships between tables, forming a **star schema**.

The following tables were included:

- FactVehicleAccident
- DimDate
- DimLocation
- DimDriver
- DimVehicle
- DimSeverity
- DimRoadEnvironment

This structure enables efficient multidimensional analysis.



Data Source View (Star Schema)

This figure illustrates the Data Source View (DSV) representing the star schema structure, where the FactVehicleAccident table is centrally connected to all dimension tables through foreign key relationships.

3.5 Cube Design

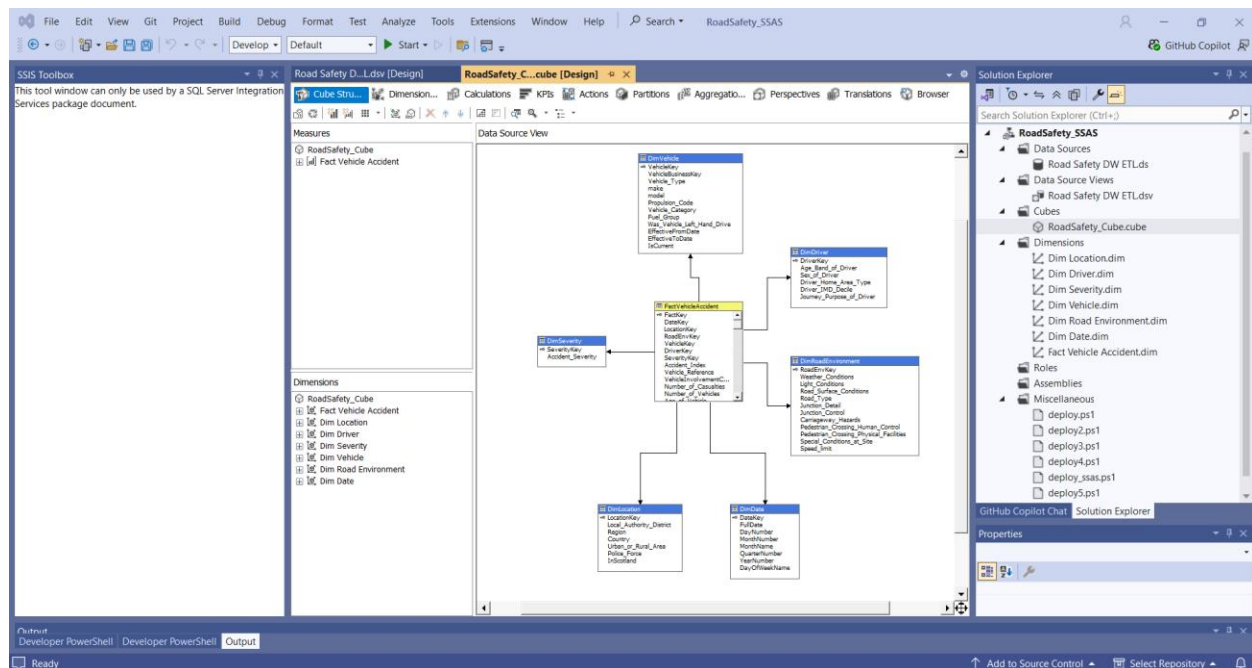
The cube was created using the Cube Wizard by selecting:

- **Measure Group:** FactVehicleAccident
- **Dimensions:** All dimension tables

The cube was named:

RoadSafety_Cube

The relationships between the fact table and dimensions were configured in the **Dimension Usage** tab.



Cube Structure Design

This figure shows the structure of the *RoadSafety_Cube*, including the measure group derived from the fact table and the associated dimensions used for multidimensional analysis.

3.6 Measures and Fact Table Integration

Measures were created from the *FactVehicleAccident* table to support analysis. These measures include:

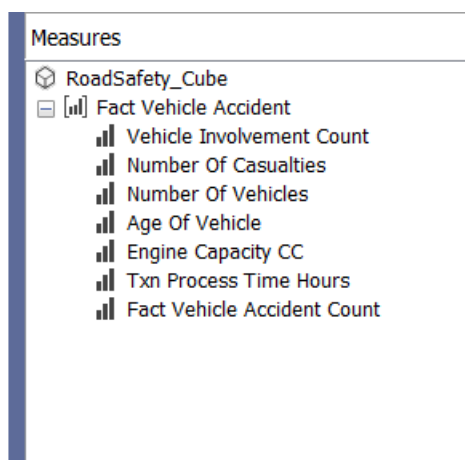
- Fact Vehicle Accident Count
- Number of Casualties
- Number of Vehicles
- Age of Vehicle
- Engine Capacity CC
- Transaction Process Time Hours

These measures allow analysis of accident trends, severity impact, and vehicle-related insights.

Measure Aggregation Logic

The following aggregation rules were applied:

Measure	Aggregation
Accident Count	SUM
Number of Casualties	SUM
Number of Vehicles	SUM
Age of Vehicle	SUM
Engine Capacity	SUM
Txn Process Time	SUM



Measures in FactVehicleAccident

This figure shows the measures defined in the FactVehicleAccident measure group, which are used to perform quantitative analysis in the OLAP cube.

3.7 Dimension Configuration

Each dimension was configured with appropriate attributes and descriptive columns.

Example: DimSeverity

The severity dimension was configured to display meaningful values:

- Fatal
- Serious
- Slight

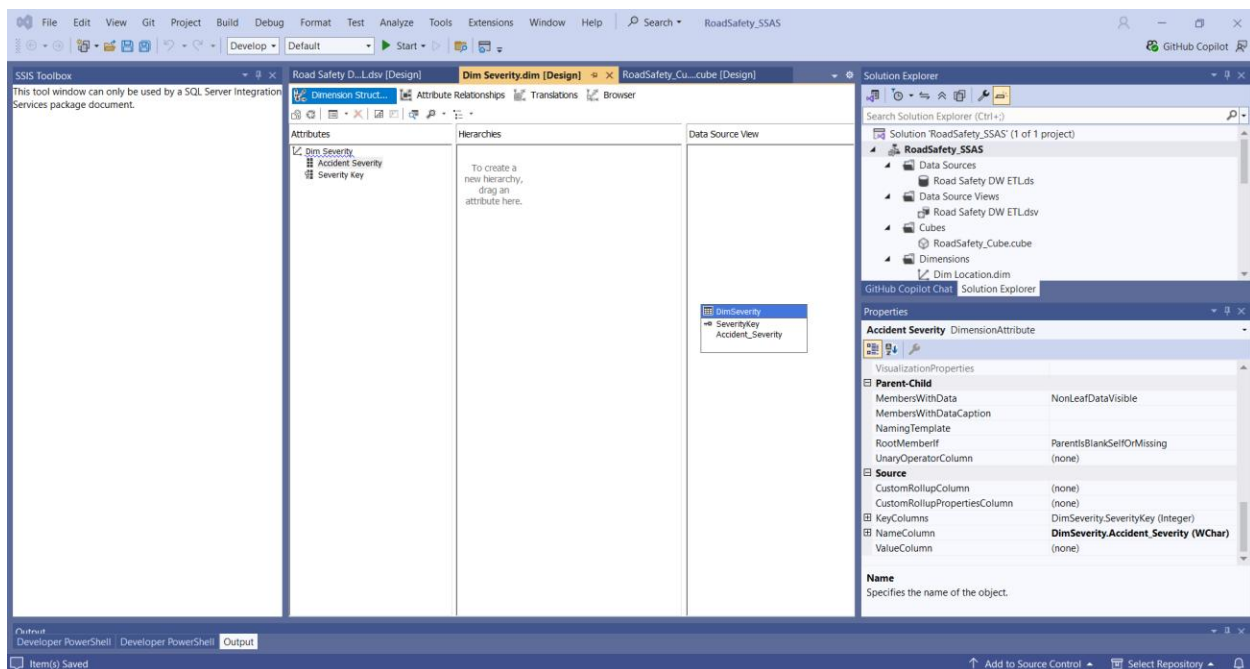
This was achieved by:

- Setting **Key Column = SeverityKey**

- Setting Name Column = Accident_Severity

Other Dimensions Configured

- **DimDate** → Date attributes
- **DimLocation** → Country, Region, District
- **DimDriver** → Driver-related attributes
- **DimVehicle** → Vehicle characteristics
- **DimRoadEnvironment** → Weather, Road Type, Light Conditions



Dimension Configuration – DimSeverity

This figure shows the configuration of the DimSeverity dimension, where the key column and name column are defined to display meaningful severity categories such as Fatal, Serious, and Slight.

3.8 Hierarchy Implementation

Hierarchies were implemented to support OLAP operations such as roll-up and drill-down.

Date Hierarchy

Year → Quarter → Month → Date

This hierarchy enables time-based analysis at different levels.

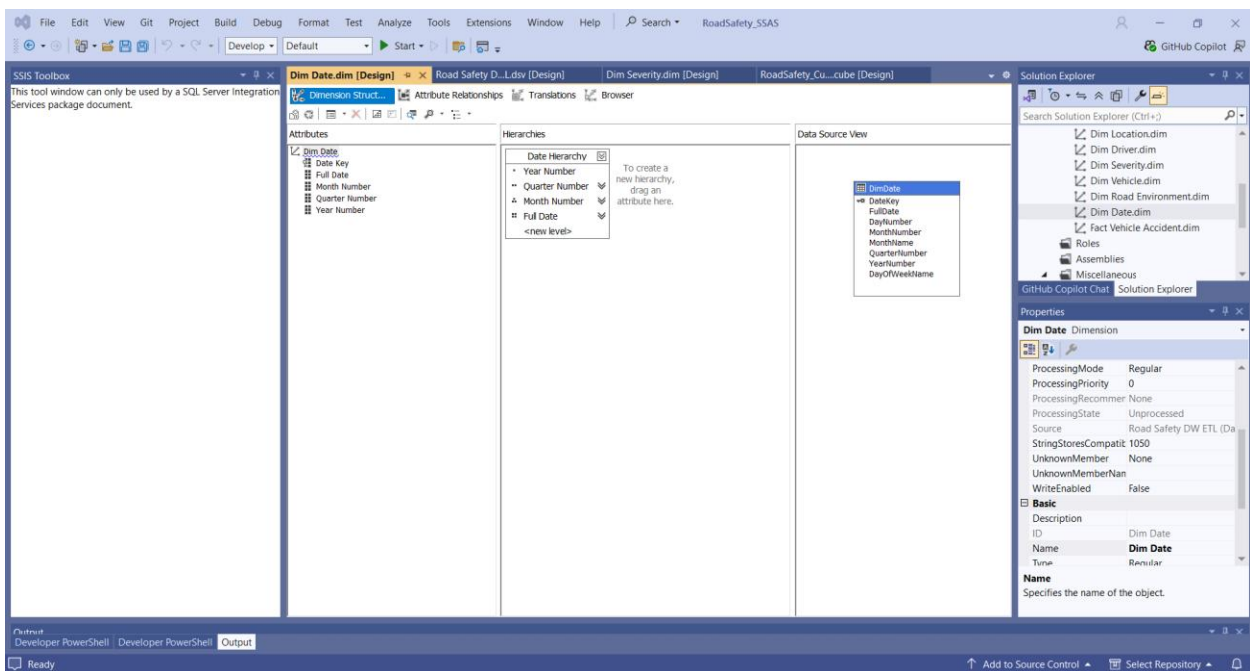
Location Hierarchy

Country → Region → Local Authority District

This hierarchy enables geographical analysis of accidents.

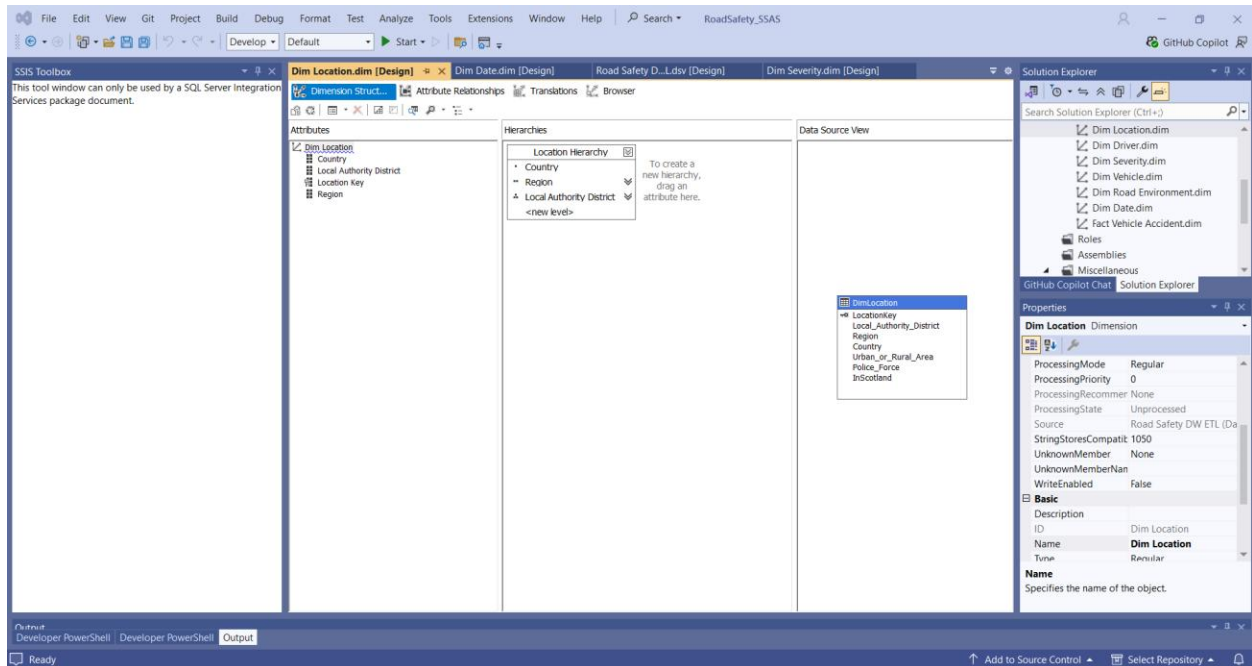
Purpose of Hierarchies

- Support drill-down analysis
- Improve query performance
- Enable structured navigation



Date Hierarchy Implementation

This figure shows the Date hierarchy, which enables drill-down and roll-up operations from year to day level.



Location Hierarchy Implementation

This figure shows the Location hierarchy, which enables geographical analysis of accident data from country level down to district level.

3.9 Cube Deployment and Processing

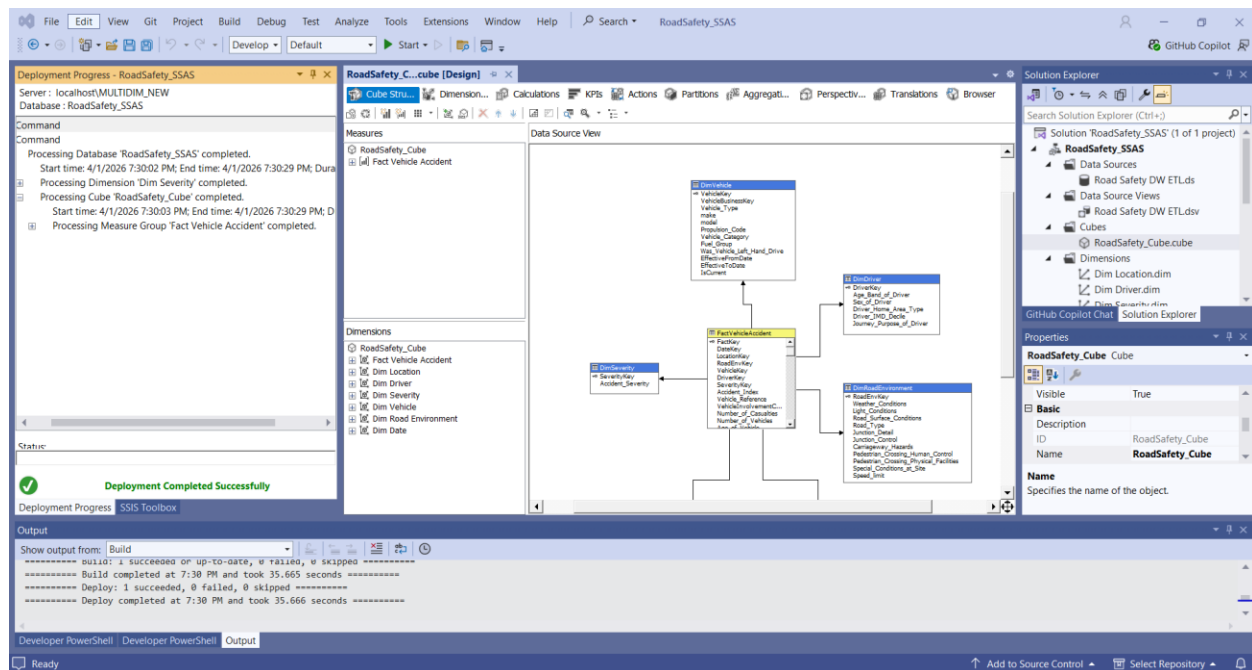
The cube was deployed to the SSAS server and processed successfully.

During deployment:

- Dimensions were processed
- Cube data was loaded
- No errors were encountered

The cube processed approximately:

1.8 million records



Cube Deployment and Processing

This figure shows the successful deployment and processing of the OLAP cube in SSAS, indicating that all dimensions and measures were loaded without errors.

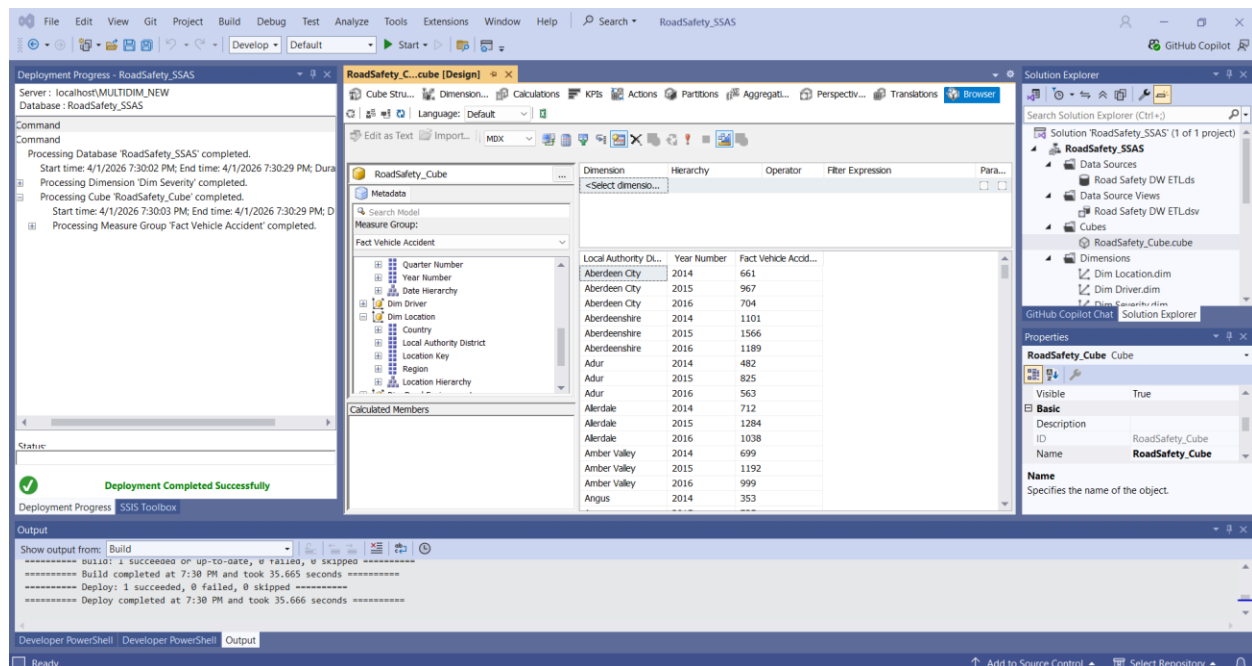
3.10 Cube Browsing and Validation

The cube was validated using the Browser tab to ensure:

- Correct aggregation
- Proper dimension relationships
- Accurate data representation

Example validations performed:

- Accident Count by Year
- Casualties by Location
- Vehicle trends over time



Cube Browser Validation

This figure shows the cube browsing interface, where accident count is analyzed by year, confirming correct aggregation and proper functioning of the OLAP cube.

4. OLAP Operations using Excel

4.1 Introduction to Excel OLAP Analysis

Online Analytical Processing (OLAP) is a technique used to analyze multidimensional data from different perspectives. In this assignment, Microsoft Excel is used as a front-end tool to interact with the SSAS cube and perform analytical operations.

Excel PivotTables provide a powerful interface to:

- Summarize large datasets
- Analyze measures across multiple dimensions
- Perform OLAP operations such as Roll-up, Drill-down, Slice, Dice, and Pivot

The analysis is conducted using the **RoadSafety_SSAS cube**, which contains key measures such as accident counts, casualties, number of vehicles, and vehicle-related attributes.

4.2 Connecting Excel to SSAS Cube

The SSAS cube was connected to Microsoft Excel using the following steps:

1. Open Excel and navigate to the **Data tab**
2. Select **Get Data** → **From Database** → **From Analysis Services**
3. Enter the server name (e.g., localhost\MULTIDIM_NEW)
4. Select the database: **RoadSafety_SSAS**
5. Choose the cube: **RoadSafety_Cube**
6. Load the cube into a PivotTable

This connection allows Excel to query and analyze cube data dynamically.

Roll-up is used to aggregate data to a higher level in a hierarchy.

Implementation:

- **Rows:** Location Hierarchy (Country → Region)
- **Columns:** Year Number
- **Values:** Accident-related measures

Result:

Data is summarized from lower levels (districts) to higher levels (regions and country). This provides a high-level overview of accident statistics.

The screenshot shows an Excel PivotTable with the following data:

Country	Region	Fact Vehicle Accident Count	Number Of Casualties	Number Of Vehicles	Age Of Vehicle	Engine Capacity CC
UK	East Midlands	2651	3617	5364	20281	7369391
	North West	5856	9597	12045	42724	12868527
	Wales	1793	2747	3802	13794	4137979
	West Midlands	7408	10961	15466	60079	17074200
	Yorkshire and the Humber	9159	13353	18916	69180	23556674
Grand Total		395221	581715	834396	3011691	938457035

Roll-Up Operation (Aggregated Data by Region and Country)

4.4 Drill-Down Operation

Drill-down allows users to move from summarized data to more detailed levels.

Implementation:

- Expanded **Location Hierarchy**:
 - Country → Region → Local Authority District
- Used **Date Hierarchy**:
 - Year → Month → Full Date

Result:

Users can analyze detailed accident data at district level and even at daily level, providing deeper insights.

Year Number	Quarter Number	Month Number	Full Date	Fact Vehicle Accident Count	Number Of Casualties	Number Of Vehicles	Age Of Vehicle	Engine Capacity CC
2014	1	February	2014-02-01					
			2014-02-02	1	1	2	7	
			2014-02-03	6	6	12	18	
			2014-02-04	16	17	49	134	
			2014-02-05	3	3	5	24	
			2014-02-06					
			2014-02-07	1	1	2	2	
			2014-02-08					
			2014-02-09	11	11	22	82	
			2014-02-10	8	8	16	51	
			2014-02-11	10	25	16	120	
			2014-02-12	5	5	5	29	
			2014-02-13	9	9	26	93	
			2014-02-14	10	13	27	64	
			2014-02-15	7	7	13	71	
			2014-02-16	3	3	5	33	
			2014-02-17	6	15	12	16	
			2014-02-18	11	15	22	140	
			2014-02-19					
			2014-02-20	10	10	24	102	
			2014-02-21	15	19	36	70	
			2014-02-22	10	16	17	88	

Drill-Down Operation (Detailed View by Date and District)

4.5 Slice Operation

Slice filters the data based on a single dimension.

Implementation:

- Applied filter:
 - **Year Number = 2014**

Result:

The PivotTable displays data only for the selected year while keeping all other dimensions intact.

Year Number	2014					
Row Labels	Fact Vehicle Accident Count	Number Of Casualties	Number Of Vehicles	Age Of Vehicle	Engine Capacity CC	Txn Process Time Hours
UK	368354	541440	778803	2805633	873450264	1841770
East Midlands						
Nottingham	2651	3617	5364	20281	7369391	13255
North West						
Liverpool	3250	5205	6656	22737	7402788	16250
Manchester	2606	4392	5389	19987	5465739	13030
Wales						
Cardiff	1793	2747	3802	13794	4137979	8965
West Midlands						
Birmingham	7408	10961	15466	60079	17074200	37040
Yorkshire and the Humber						
Leeds	5513	7887	11215	40479	14955634	27565
Sheffield	3646	5466	7701	28701	8601040	18230
Grand Total	395221	581715	834396	3011691	938457035	1976105

Slice Operation (Filtered by Year 2014)

4.6 Dice Operation

Dice filters data using multiple dimensions simultaneously.

Implementation:

Applied filters:

- **Year Number** (e.g., 2015)
- **Region** (e.g., North West)
- **Weather Conditions** (Fine, no high winds)
- **Accident Severity** (Fatal)

Result:

A focused subset of data is displayed, allowing multi-dimensional analysis.

The screenshot shows an Excel PivotTable with the following data:

Row Labels	Fact Vehicle Accident Count	Number Of Casualties	Number Of Vehicles	Age Of Vehicle	Engine Capacity CC	Txn Process Time Hours	Total Fact Vehicle Accident Count
UK							
North West							
Liverpool	27	29	36	185	95792	135	27
Manchester	33	47	56	267	19781	165	33
Grand Total	60	76	92	452	115573	300	60

Dice Operation (Multiple Filters Applied)

4.7 Pivot Operation

Pivot changes the orientation of the data by swapping rows and columns.

Implementation:

- **Rows:** Year Number
- **Columns:** Region
- **Values:** Accident-related measures

Result:

This enables comparison of accident statistics across different regions for each year.

PivotTable Fields

Choose fields to add to report:

Search

Fact Vehicle Accident
☒ Age Of Vehicle
☒ Engine Capacity CC
☒ Fact Vehicle Accident Count
☒ Number Of Casualties
☒ Number Of Vehicles
☒ Txn Process Time Hours

Drag fields between areas below:

Filters

Weather Conditions
 Accident Severity

Columns

Region

Rows

Year Number

Values

Fact Vehicle Accident...
 Number Of Casualties
 Number Of Vehicles

☐ Defer Layout Update

Update

Pivot Operation (Year vs Region Analysis)

4.8 Summary of OLAP Analysis

The OLAP operations performed using Excel demonstrated the flexibility and effectiveness of multidimensional analysis.

Key observations:

- Roll-up provided aggregated summaries for high-level insights
- Drill-down enabled detailed analysis at district and date level
- Slice allowed focused analysis on a single dimension
- Dice enabled complex filtering across multiple dimensions
- Pivot changed the data perspective for better comparison

Overall, Excel combined with SSAS provided a powerful analytical environment for exploring and understanding road safety data.

5. Power BI Reports Development

5.1 Introduction to Power BI

Microsoft Power BI is a powerful business intelligence tool used to visualize data, create interactive dashboards, and generate analytical reports. It allows users to connect to multiple data sources, transform data, and present insights through various visualizations such as charts, tables, and maps.

In this assignment, Power BI is used as a front-end reporting tool connected to the SSAS cube developed earlier. The reports provide interactive analysis of road accident data using filters, hierarchies, drill-down, and drill-through capabilities.

The objective of this section is to develop four different types of reports that demonstrate key BI concepts such as tabular analysis, cascading filters, hierarchical exploration, and navigation between summary and detailed views.

5.2 Data Connection and Preparation

The Power BI reports were created using a **Live Connection to the SSAS Cube (RoadSafety_Cube)**.

Steps followed:

1. Open Power BI Desktop
2. Navigate to:
Home → Get Data → SQL Server Analysis Services
3. Enter:
 - Server Name: localhost\MULTIDIM_NEW
 - Database: RoadSafety_SSAS
4. Select:
 - Cube: RoadSafety_Cube
5. Choose:
Connect Live

Data Preparation:

Since a live connection was used:

- No data transformation was required
- Measures and hierarchies were already defined in SSAS
- Dimensions and attributes were directly available

Available Data:

The following elements were used:

- Fact Measures:
 - Fact Vehicle Accident Count
 - Number of Casualties
 - Number of Vehicles
- Dimensions:
 - Date (Year, Quarter, Month)
 - Location (Country, Region, District)
 - Severity
 - Road Environment
 - Vehicle attributes

5.3 Data Modeling and Relationships

In this implementation, data modeling was handled in SSAS using a **star schema**.

Key Components:

- Fact Table:
 - FactVehicleAccident
- Dimension Tables:
 - DimDate
 - DimLocation
 - DimSeverity
 - DimVehicle
 - DimDriver

- DimRoadEnvironment

Relationships:

- One-to-many relationships between dimensions and fact table
- Dimensions linked via foreign keys
- Optimized for analytical queries

Hierarchies Used:

- Date Hierarchy:
 - Year → Quarter → Month
- Location Hierarchy:
 - Country → Region → Local Authority District

These hierarchies were directly used in Power BI for drill-down and grouping.

5.4 Report 1 – Matrix Report

Objective:

To display detailed tabular data with row and column groupings using a Matrix visual.

Implementation:

A matrix visual was created to analyze accident data across location and time dimensions.

Configuration:**Rows:**

- Location Hierarchy
 - Country → Region → Local Authority District

Columns:

- Date Hierarchy

- Year → Quarter → Month

Values:

- Fact Vehicle Accident Count
- Number of Casualties
- Number of Vehicles

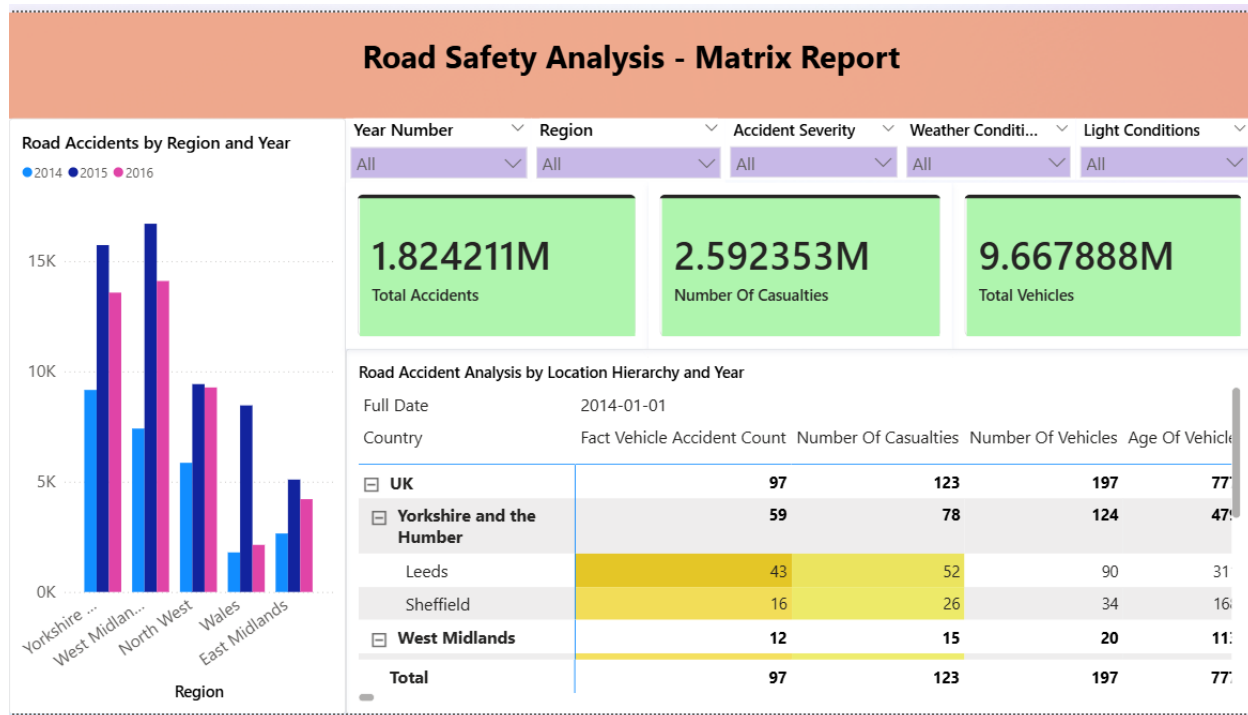
Additional Features:

- KPI Cards added:
 - Total Accidents
 - Total Casualties
 - Total Vehicles
- Conditional formatting applied to highlight high values
- Filters (slicers) added:
 - Year Number
 - Region
 - Severity
 - Weather Conditions
 - Light Conditions

Outcome:

This report enables:

- Multi-dimensional tabular analysis
- Comparison across regions and time
- Hierarchical navigation within the matrix



5.5 Report 2 – Slicer-Based Report (Cascading Filters)

Objective:

To implement multiple slicers where one slicer dynamically filters another (cascading filters).

Implementation:

Multiple slicers were added to filter data interactively.

Slicers Used:

- Region
- District (Local Authority District)
- Accident Severity
- Weather Conditions
- Light Conditions

Cascading Filter Logic:

- Selecting a **Region** automatically filters the available **Districts**
- This creates a dependency between slicers

Visuals Used:

- Line Chart:
 - Casualties Trend Over Time
- Bar Chart:
 - Accident Count by Region and Year
- Pie Chart:
 - Accident Severity Distribution
- KPI Cards:
 - Total Accidents
 - Total Casualties
 - Total Vehicles

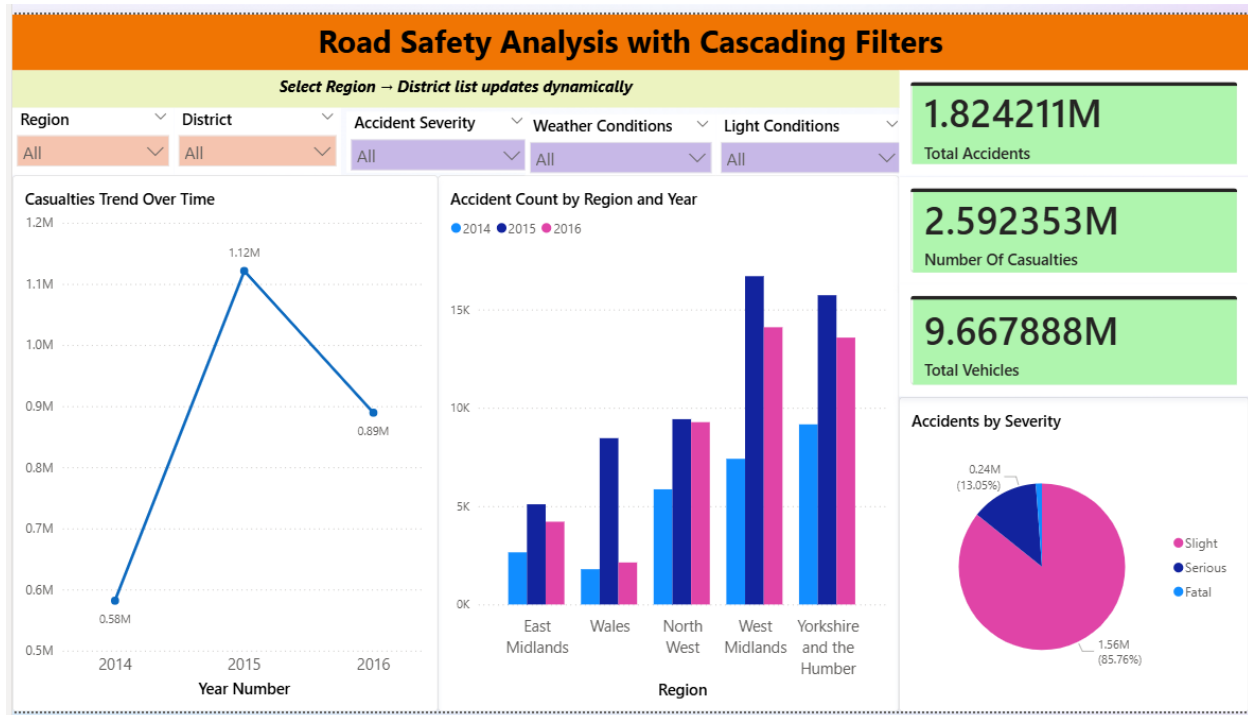
Features:

- Cross-filtering enabled between visuals
- Tooltips added (Age of Vehicle, Engine Capacity)
- Dynamic updates based on slicer selections

Outcome:

This report demonstrates:

- Interactive filtering
- Cascading slicer behavior
- Multi-perspective analysis



5.6 Report 3 – Drill-Down Report

Objective:

To allow users to explore data hierarchically using drill-down functionality.

Implementation:

Charts were created using the Date hierarchy.

Hierarchy Used:

- Year → Quarter → Month

Visuals:

- Bar Chart:
 - Accident Count by Year → Quarter → Month
- Line Chart:
 - Casualties Trend across hierarchy

- KPI Cards:
 - Total Accidents
 - Total Casualties
 - Total Vehicles

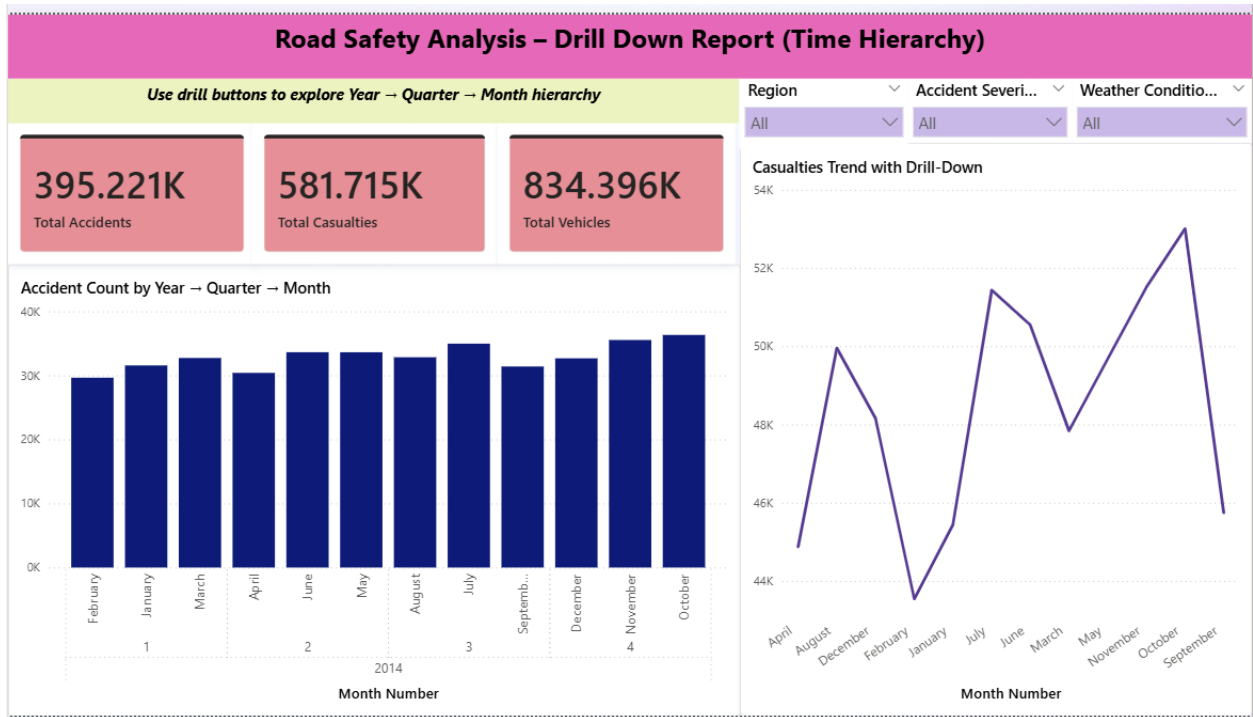
Features:

- Drill-down buttons enabled
- Users can:
 - Expand hierarchy
 - Navigate between levels
- Month sorting corrected using Month Number
- Instruction text added for usability

Outcome:

This report allows:

- Detailed time-based analysis
- Hierarchical navigation
- Trend exploration



5.7 Report 4 – Drill-Through Report

Objective:

To enable navigation from a summary report to a detailed report using drill-through.

Implementation:

Two pages were created:

1. Main Page (Summary Dashboard)

Visuals:

- KPI Cards
- Bar Chart (Accidents by Region)
- Pie Chart (Severity Distribution)
- Line Chart (Trend Over Time)

Instruction
“Right-click a region to view detailed analysis”

added:

2. Detailed Page

Configured as drill-through target.

Drill-Through Setup:

- Field used:
 - Region
- Added
Drill-through filters panel

to:

Visuals in Detailed Page:

- KPI Cards
- Line Chart (Casualties by Date)
- Bar Chart (District-level accidents)
- Pie Chart (Severity distribution)
- Matrix (District-level breakdown)
- Top N Chart:
 - Top 5 Districts by Accident Count

Additional Features:

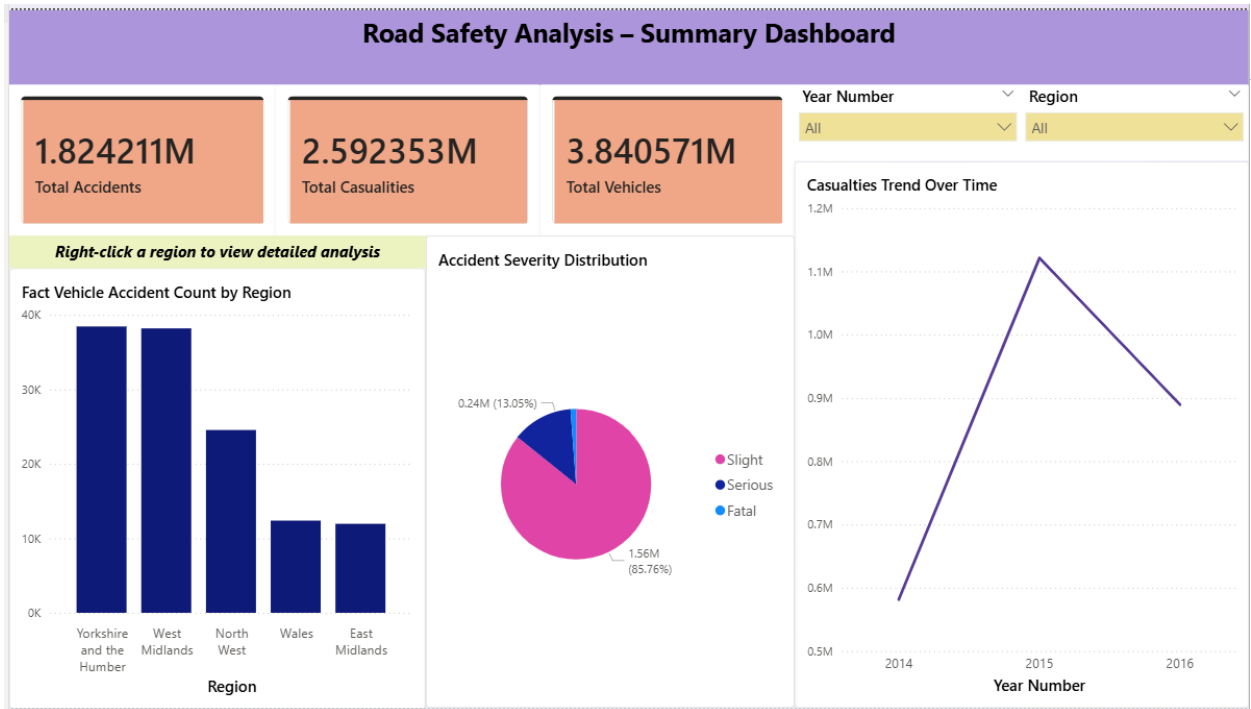
- Back button added for navigation
- Cross-filtering enabled
- Tooltips added

Outcome:

This report demonstrates:

- Navigation between summary and detail

- Context-based filtering
- Deep analytical exploration



5.8 DAX Calculations (if used)

Although the project primarily uses SSAS measures, additional calculations were implemented.

Example Measure:

Avg Casualties per Accident =
DIVIDE([Number Of Casualties], [Fact Vehicle Accident Count])

Purpose:

- To calculate average impact per accident
- Avoid division by zero errors

Usage:

- Added to matrix visual
- Used for analytical insights

5.9 Report Publishing (Power BI Service)

Steps:

1. Click:
Home → Publish
2. Select:
 - Workspace
3. Upload report to Power BI Service

Verification:

- Open report in browser
- Ensure:
 - All visuals load correctly
 - Filters work
 - Drill-down and drill-through function properly

Demonstration Requirement:

Reports were demonstrated by:

- Opening in Power BI Service (online)
- Showing interactivity and navigation

5.10 Summary of Reports

The Power BI reports successfully demonstrate various business intelligence capabilities.

Key Achievements:

- Matrix report for tabular analysis
- Cascading slicers for dynamic filtering
- Drill-down for hierarchical exploration
- Drill-through for detailed navigation

Benefits:

- Interactive dashboards
- Multi-dimensional analysis
- Improved decision-making

Conclusion:

The integration of SSAS and Power BI enabled the transformation of structured data into meaningful insights, supporting advanced analytics in road safety analysis.